

314707003_2.25.R

CDH

2026-03-09

```
rm(list=ls())

#install.packages("devtools")
#library(devtools)
#install_github("ccolonescu/PoEdata")
#library(PoEdata)
#####

#problem 2.25
url <- "http://www.principlesofeconometrics.com/poe5/data/rdata/cex5_small.rdata"
dest <- file.path(tempdir(), "cex5_small.rdata")

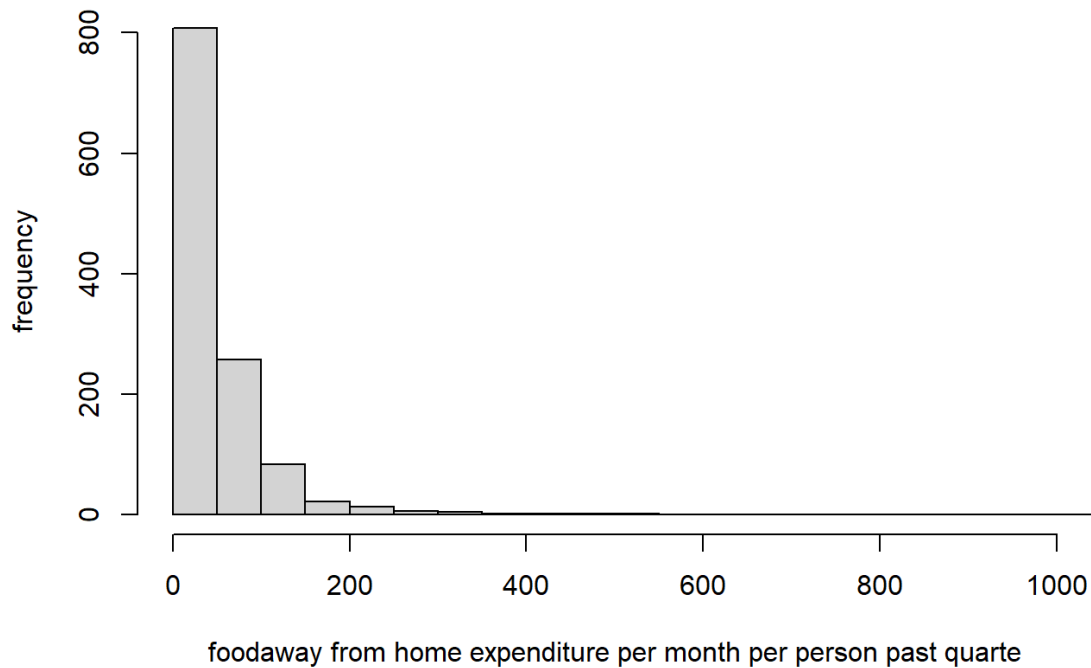
download.file(url, dest, mode = "wb")
load(dest)

#(a)
cat(
  "Mean of FOODWAY is:", mean(cex5_small$foodaway), "\n",
  "Median of FOODWAY is:", median(cex5_small$foodaway), "\n",
  "25th percentiles of FOODWAY is:", quantile(cex5_small$foodaway, 0.25),
  "\n",
  "75th percentiles of FOODWAY is:", quantile(cex5_small$foodaway, 0.75),
  "\n",
  sep = " "
)
```

```
## Mean of FOODWAY is: 49.27085
## Median of FOODWAY is: 32.555
## 25th percentiles of FOODWAY is: 12.04
## 75th percentiles of FOODWAY is: 67.5025
```

```
hist(cex5_small$foodaway,xlab = 'foodaway from home expenditure per month per
person past quarte', ylab = 'frequency', main = 'Histogram of FOODAWAY',
      xlim = c(0,1000), breaks= 20)
```

Histogram of FOODAWAY

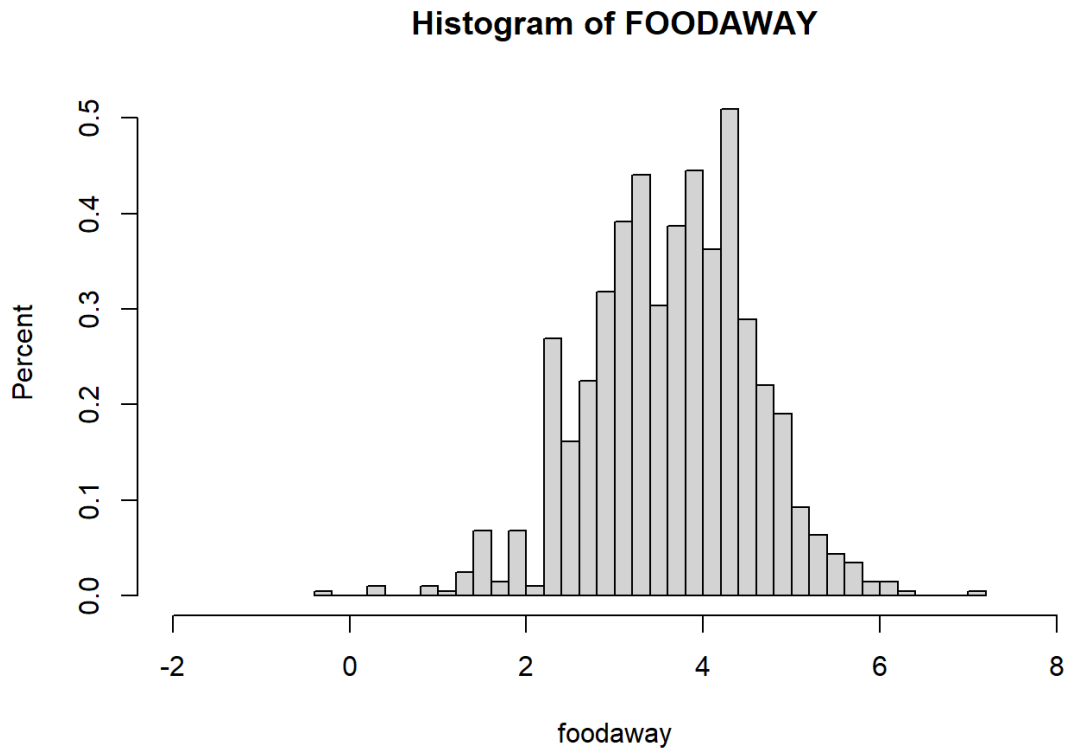


```
#(b)
cat(
  "advanced = 1:", "\n",
  "N:", sum(cex5_small$advanced), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$advanced == 1]), "\n",
  "Median:", median(cex5_small$foodaway[cex5_small$advanced == 1]), "\n",
  "college = 1:", "\n",
  "N:", sum(cex5_small$college), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$college == 1]), "\n",
  "Median:", median(cex5_small$foodaway[cex5_small$college == 1]), "\n",
  "None:", "\n",
  "N:", sum(cex5_small$advanced == 0 & cex5_small$college == 0), "\n",
  "Mean:", mean(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0]), "\n",
  "Median:", median(cex5_small$foodaway[cex5_small$advanced == 0 & cex5_small$college == 0]), "\n",
  sep = ""
)
```

```
## advanced = 1:
## N:257
## Mean:73.15494
## Median:48.15
## college = 1:
## N:369
## Mean:48.59718
## Median:36.11
## None:
```

```
## N:574
## Mean:39.01017
## Median:26.02
```

```
##(c)
hist(log(cex5_small$foodaway),prob = TRUE,xlab = 'foodaway', ylab = 'Percent', main = 'Histogram of FOODAWAY',
      xlim = c(-2,8), breaks= 30)
```



```
cat(
  "number of values loss after take log is:",sum(!is.finite(log(cex5_small
    $foodaway))), "\n",
  "since those sample are 0,and ln(0) is undefined and become missing value",
  sep = ""
)
```

```
## number of values loss after take log is:178
## since those sample are 0,and ln(0) is undefined and become missing value
```

```
##(d)
idx <- !is.na(cex5_small$foodaway) & cex5_small$foodaway > 0 &
  !is.na(cex5_small$income)

df <- cex5_small[idx, ]
```

```

mod1 <- lm(log(foodaway) ~ income, data = df)
intercept <- coef(mod1)[1]
slope <- coef(mod1)[2]
cat(
  "intercept is: ", intercept, "\n",
  "slope is: ", slope, "\n",
  "We estimate that each additional $100 household income increases food aw
ay expenditures", "\n",
  "per person of about 0.69%, other factors held constant", "\n",
  sep=""
)

```

```

## intercept is: 3.1293
## slope is: 0.006901748
## We estimate that each additional $100 household income increases food away
expenditures
## per person of about 0.69%, other factors held constant

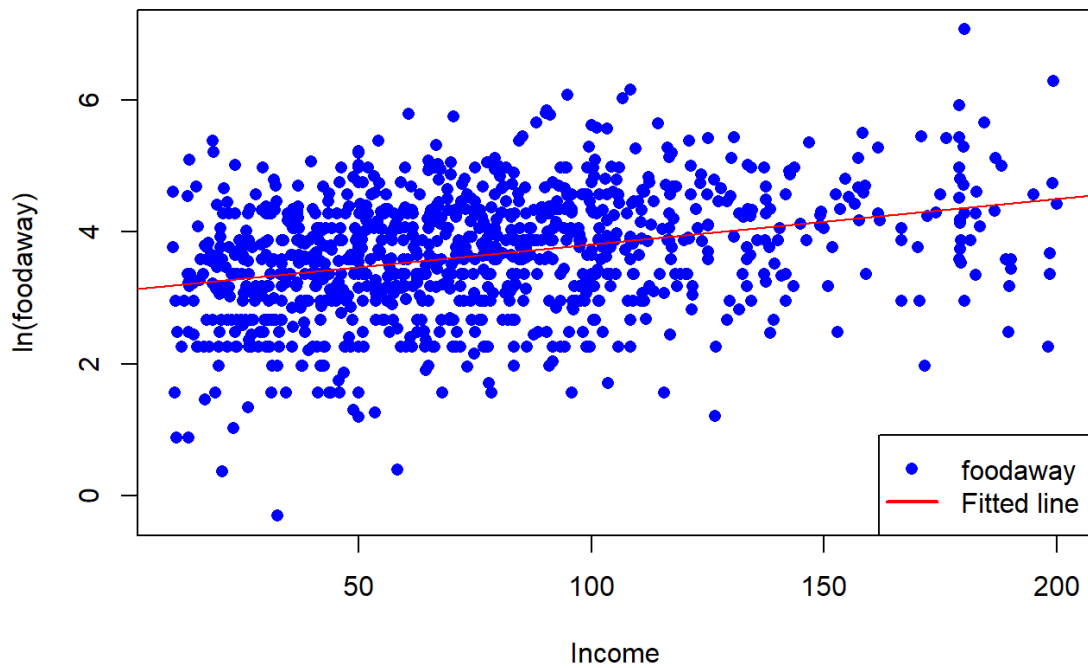
```

```

#(e)
lnfood = log(cex5_small$foodaway)
plot(cex5_small$income, lnfood,
     xlab="Income",
     ylab="ln(foodaway)",
     main = "obs and log-linear fitted line",
     pch = 16, col = 'blue',
     type = "p")
abline(intercept, slope, col = 'red')
legend("bottomright", legend = c("foodaway", "Fitted line"),
     col = c("blue", "red"), pch = c(16, NA), lty = c(NA, 1),
     lwd = c(NA, 2))

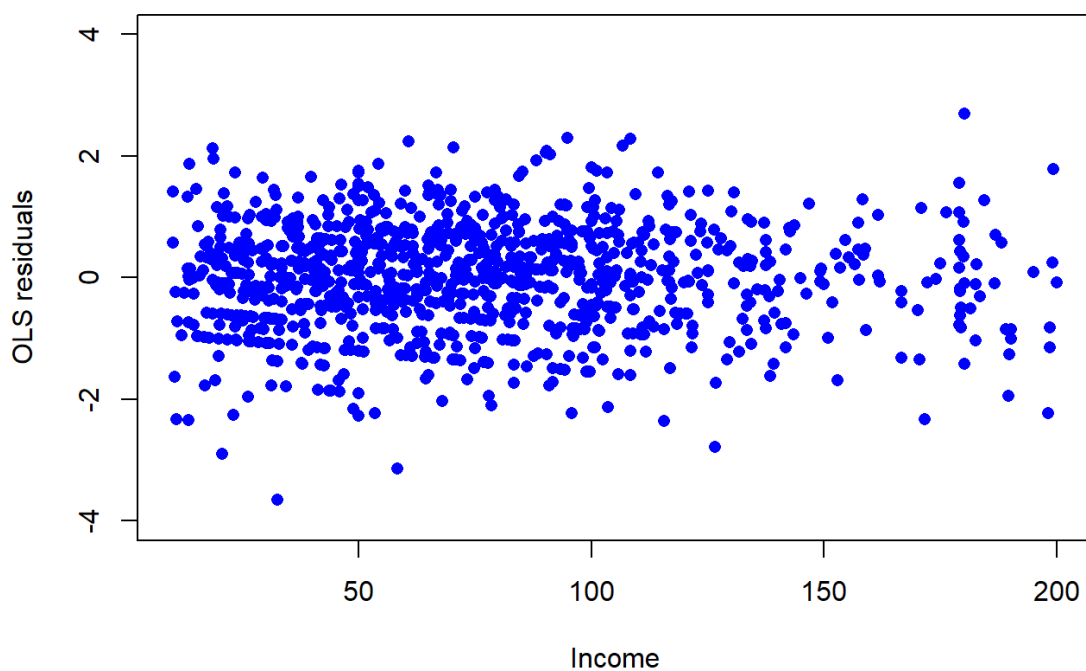
```

obs and log-linear fitted line



```
##(f)
res = resid(mod1)
plot(df$income,res,
      xlab = "Income",ylab = "OLS residuals",
      main = "Residual vs. Income",
      ylim = c(-4,4),
      pch =16,col = 'blue',
      type='p')
```

Residual vs. Income



```
cat(  
  "The OLS residuals do appear randomly distributed with no obvious patterns.  
  There are fewer", "\n",  
  "observations at higher incomes, so there is more \"white space.\""  
  sep = ""  
)
```

```
## The OLS residuals do appear randomly distributed with no obvious patterns.  
There are fewer  
## observations at higher incomes, so there is more "white space."
```